An Intelligent Interactive Learning Platform for Children with Dyscalculia Using Machine Learning and Adaptive Learning Techniques

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***Abstract*—An interactive learning tool combining machine learning and adaptive learning techniques to aid children with dyscalculia learning mathematics is presented. The system consists of four system components, namely: cognitive skill profiling, personalized mathematics learning, adaptive game-based learning, and learning insight and early warning. Data on the interaction of the learner is processed to determine their cognitive characteristics, to estimate the level of skills acquired, to adjust the level of difficulty of the game and to find risks for learning. The average Macro F1 was 0.640 in the Cognitive Skill Profiling System with the selected Random Forest model. With the chosen rule-based model, the Adaptive Game Based Learning Engine was implemented with a 55.6% completion rate, 61.2% challenge-fit, 58.8% engagement, and 72.4% safety and stability. Logistic Regression had a Balanced Accuracy of 47.7% and a F2 Score of 0.689 with the Learning Insight and Early Warning System. Bayesian Knowledge Tracing, Reinforcement Learning, clustering and predictive analytics are integrated throughout the system to enable personalized learning and ongoing learner monitoring. The proposed platform will offer personalized mathematical activity, interactive games, adaptive difficulty, engagement-based learning strategies and insights into progress within a single learning environment.**

***Main keywords— Dyscalculia, Personalized Learning, Adaptive Learning, Game-Based Learning, Machine Learning, Learning Analytics, Reinforcement Learning*.**

I. INTRODUCTION

Although mathematics is crucial to academic success and everyday decision making, some children with developmental dyscalculia might have ongoing problems understanding numbers, counting, comparing numbers in terms of size and magnitude, and performing arithmetic skills. These struggles can impact academic achievement, self-esteem and motivation. Digital learning technologies offer the potential for interactive and adaptive mathematics learning, but most current digital learning systems are based on predetermined activities and basic mathematics scores, and lack attention to learner behavior, cognition, engagement, and learning progress. It is therefore possible for learners with different difficulties to experience the same learning experience, but different learning needs. To overcome this drawback, this research introduces an Interactive Learning Tool for children with Dyscalculia with four interlaced components: Cognitive Skill Profiling, Personalized Math Learning, Adaptive Game-Based Learning, and Learning Insight and Early Warning. The system uses learning interactions to discover cognitive characteristics, predict mastery level based on Bayesian Knowledge Tracing, increase or decrease game difficulty and rewards based on Reinforcement Learning, and track learning progress as well as potential problems with learning using Learning Analytics. The proposed platform features these elements to form a continuous learning cycle where the interactions of learners are analyzed to personalize learning activities in the next step. The system should offer a more individualized and engaging mathematics education experience for children with dyscalculia, and meaningful mathematics information for teachers and parents for progress monitoring and timely intervention.

II. LITERATURE REVIEW

Developmental dyscalculia is a specific learning difficulty with numerical understanding, arithmetic performance and mathematical reasoning. Common characteristics of dyscalculia are difficulties in number sense, working memory, the ability to sustain attention, processing speed, and also numerical relationships [1]–[3]. These challenges can have a negative impact on school performance and self-esteem, and can result in early intervention if identified and properly addressed [1], [2]. Learning difficulties identification and Dyscalculia screening has been previously studied using AI and machine learning. Machine learning

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models like Decision Trees, Random Forest, Gradient Boosting, and other classification models [4]– [7], [25] and [26] were used to analyse learning performance and learning behaviours. Educational data mining has also proved the benefits of analysing learner interaction data such as accuracy, response time, attempts and degree of engagement to understand learning behaviour and aid in the education decision-making process [8, 27, 40].

Digital learning applications have increasingly helped children with math learning disabilities, as have serious games. A few programs provide the number comparison activities, numerical representation and arithmetic practice such as The Number Race [11] or Calcularis [12]. Other studies have examined the use of learning environment for math understanding and motivation of learners through games [4], mobile application [9] and [10] and augmented reality [13] and [14] based learning robotics. Gamification tools such as points, levels, reward, badges and challenges are used to encourage learners to practise and keeping them motivated in learning [14, 16]. Another important point emphasized by Universal Design for Learning is providing alternative ways to access content as well as information that has been grasped and alternative means to demonstrate what the learner has learned. Some of the existing applications are oriented towards specific mathematical skills or tasks and have not yet been able to assess continuously learners' cognitive capacities, behavioral patterns, and overall development [7] [9] and [13].

Personalized learning system is a system that provides a learning activity based on the individual learner's knowledge, performance and needs. Intelligent tutoring systems (ITS) and adaptive mathematics applications have used learner models to determine what and how much a learner knows to select learning activities for them [8, 17]. A widely-used way to estimate the probability of a learner having learnt a certain skill and adjust the probability in response to the learner's action is called Bayesian Knowledge Tracing (BKT) [18, 19, 28, 36]. The ability of Deep Knowledge Tracing has also been extended to model complex learning patterns from a series of learner interactions [18]. The approaches allow for personalized learning, but are mainly about accuracy of answers rather than cognitive aspects, engagement or behaviour patterns [17, 19]. Hence, an integrated learner model is required which is able to connect cognitive profiling with continuous estimation of mastery and generation of learning-paths that match the learner.

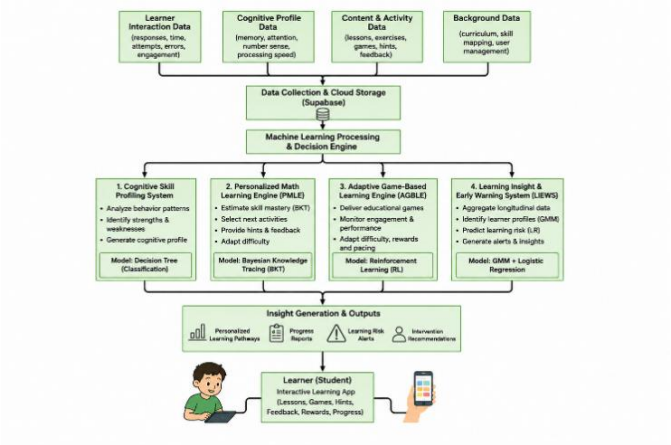
Creating a motivational context for pupils to practise mathematical skills, through an interactive, engaging and rewarding task implies game-based learning [9, 10, 14, 16]. However, the difficulty of activities and preset sequences of activities may not be appropriate for the needs of individual learners. Reinforcement Learning (RL) is a technique that selects an action based on the feedback received from the learner after performing an action and from a current state of the learner [20, 29, 37]. RL can be implemented to change the difficulty, learning rate, content, hints, rewards, and progression in educational setting according to the learner's performance [20, 21, 30, 38]. RL has also been studied for instructional sequencing [20] and adaptive educational environments [21] as well as dynamic DA [30] and [31]. In some cases, however, the RL-based systems may be difficult to implement when enough interaction data and reward functions need to be provided to ensure that the actions that are selected improve learning rather than merely engaging the learner. Rule-based adaptation and RL can provide understandable and controlled adaptations, and allow for more flexible decision making [16],[20],[21].

Learning analytics and early-warning systems have been created to detect learner-performance patterns and students needing extra assistance. Typical systems include learning difficulty or academic risk prediction systems that are based on academic performance, engagement, attendance and interaction data [32], [40]–[43] . The grouping of learners based on the similarities in their performance and behavioural traits can be achieved using a clustering technique like K-Means [34]. One clustering method is the Gaussian Mixture Models (GMMs), which can model overlapping clusters of learners and also estimate the probability of membership of every learner to different clusters. Additionally, some predictive models such as Logistic Regression, Decision Trees, Random Forests have been used for student at risk detection and early intervention [25, 32, 41, 42, 43]. Usually, these models are assessed through the Accuracy, Precision, Recall, F1 score, Confusion Matrix, and AUC-ROC. Yet, a number of early-warning systems are based on generic academic risk and are not associated with detailed mathematical levels of difficulty and personalized learning activities.

In recent years, there has also been research on the use of Intelligent Tutoring Systems, Large Language Models, and inclusive educational applications in providing personalized explanations, feedback, and learning support [22]–[24] . The methods shown demonstrate the potential of AI in improving the accessibility and engagement of learners, but must be thoroughly evaluated to ensure that the content generated to support learning is accurate, appropriate, and tailored to the learning objectives [22] [23]. Moreover, most of the current systems focus on solving cognitive assessment, personalized learning, adaptive gaming and learning analytics issues in an isolated manner instead of delivering learning analytics as an integrated system. Hence, there is a research gap to create a unified system for continuous learner cognitive skill profiling, mathematical mastery estimation, activity adaptation and learning risk prediction in the context of game based learning. To address this, the proposed system integrates four types of machine learning: Decision Tree for cognitive skill profiling, Random Forest and Gradient Boosting models for personalized mathematics learning, rule based adaptive learning and Reinforcement Learning for adaptive game-based learning, K-Means, Gaussian Mixture Models and Logistic Regression for learning analytics and early-warning prediction [5, 6, 18–21, 28–39, 41–45]. This is a holistic approach that aims to provide individualized, personalized mathematical learning for the student individual and meaningful insights for teachers, parents and educational specialists.

III. METHODOLOGY

The paper presented an intelligent learning system for children with dyscalculia that adapts to the child and provides personalized mathematics learning. The proposed system



integrates four major components namely Cognitive Skill Profiling System, Personalized Math Learning Engine (PMLE), Adaptive Game-Based Learning Engine (AGBLE) and Learning Insight and Early Warning System (LIEWS). The methodology included data collection, preprocessing, development of standard models for machine learning, system integration and evaluation.

The proposed system is designed to be integrated and comprised of the Learner Application, Backend Services, Centralized Data Storage, Machine Learning Layer, and Analytics Interfaces. In learning sessions, the student uses the mobile application to interact with the mathematical exercises and game-based activities. The correctness, reaction time, number of attempts, error patterns, and engagement behaviour data are collected and sent back to the backend for further processing and storage.

The Cognitive Skill Profiling System (CSP) assesses how learners interact with the content to determine cognitive traits such as memory, attention, number sense and processing speed. The personalized learning component can use its generated cognitive profile to make decisions regarding individualized learning. The PMLE uses Bayesian Knowledge Tracing (BKT) to model the mathematical skills the learner possesses and to determine suitable learning activities to be used. The AGBLE employs Reinforcement Learning (RL) to dynamically adjust game difficulty, content, pacing and reward strategies based on learner performance and engagement. Lastly, the LIEWS summarizes the longitudinal data created by the learning components, which are used to determine learner profiles and to predict possible learning risks.

This can then be expressed as the overall data flow:

Learner Interaction→Data Collection→cognitive profiling / knowledge estimation→adaptive learning and games →longitudinal analytics – risk prediction → intervention.

This structure fosters an ongoing cycle of learning, where students' actions affect later tasks and the feedback information from all the interactions can be analysed over long periods of time.

A. Data Collection and Preprocessing

Data was collected from learners at the Sound Mind Academy, Koswaththa during mathematics learning activities and game-based learning sessions. The data collected involved learners ' responses, correctness, response time, number of attempts, error patterns, activity progression, and engagement-related information .

Learning activities involved number recognition, counting, magnitude comparison and basic calculations. Prior to the development of the model, the data was pre-processed by addressing missing or inconsistent records, removing duplicate entries, and extracting relevant features. Features such as accuracy, response time, attempt frequency, error patterns, progression and engagement were used in machine learning model development and analysis. All data from learners were anonymized and held in confidence.

B. Cognitive Skill Profiling

The Cognitive Skill Profiling System was developed to analyze learner interaction patterns and to identify cognitive characteristics associated with mathematical learning. The system addressed four cognitive dimensions: memory, attention, number sense and processing speed. Response accuracy, response time, number of attempts and behavioral patterns were used as inputs for supervised machine learning algorithms in the form of learner interaction features.

Three models have been developed and tested: Decision Tree, Random Forest and Gradient Boosting. The Decision Tree was used as a baseline model and generated interpretable classification rules from learner features. Random Forest consisted of a large number of decision tree models to increase the robustness of the classification, whereas Gradient Boosting built decision tree models sequentially to reduce the errors of the predictions. The models were trained and evaluated on the cognitive skill dataset. The evaluation metrics used were Accuracy, Precision, Recall, and Macro F1 score.

C. Personalized Mathematics Learning

The Personalized Math Learning Engine employs Bayesian Knowledge Tracing to estimate mastery of mathematical skills of learners. The knowledge state of the learner is represented by a probability. The knowledge state is updated after every learning interaction, based on the response of the learner.

The BKT model takes into account the initial knowledge of the learner, the learning probability, the guessing probability and the slipping probability. With the updated mastery level, the system suggests appropriate learning activities and, if needed, further practice or support. This allows the learning process to be tailored to the individual progress of the learner.

D. Adaptive Game-Based Learning

The Adaptive Game-Based Learning Engine combines a Rule-Based Model with Reinforcement Learning (RL) to adapt game-based learning activities dynamically based on learners' performance and behavior. Information such as accuracy, response time, number of attempts, current difficulty level, progression and engagement help to determine the state of the learner

The Rule Based Model is based on rules that prescribe appropriate adaptations according to the learner's performance. For instance, if a learner continues to get answers wrong or takes longer to answer a question the system may lower the level of the game or give more help. Once the learner masters the task repeatedly, progression of difficulty can be used.

Furthermore, the Reinforcement Learning model takes into account the state of the learner to choose the adaptation action. These may involve making the game easier or more difficult, varying the pace at which the game is played, altering the game content or changing the rules for rewards. The feedback comes in the form of the learner's performance, which is rewarded, allowing the model to adjust its adaptation over time.

The adaptive process is the sequence of:
Learner State → Adaptation Decision → Learner Response → Feedback/Reward → Updated Adaptation Strategy

Gamification features like levels, points, rewards, badges, and features to track learners' progression were added to foster engagement.

E. Learning Insight and Early Warning

The Learning Insight and Early Warning System process is an analysis of learner performance and behavior across learning sessions. Longitudinal features were extracted to detect changes in learner progress, such as accuracy trends, changes in response time, repeated errors, changes in mastery, learner stagnation, and changes in engagement.

The clustering methods used were K-Means as a baseline method and Gaussian Mixture Model (GMM) to classify the groups of learners into ones that have similar learning and behavioral patterns. The Silhouette Score and Davies–Bouldin Index were used to measure the clustering performance.

In the case of prediction of learning risk, the system employs the main machine learning technique which is Logistic Regression. Logistic Regression model uses the longitudinal learner characteristics to predict learning-risk levels. As a benchmark decision tree and random forest models were also tested for evaluation. The performance of the models was assessed by Accuracy, Precision, Recall, Macro F1-score, Confusion Matrix and AUC-ROC.

F. Model Evaluation

The developed models were assessed based on the goals of each system component. Cognitive Skill Profiling System: This model was assessed by using Accuracy, Precision, Recall and Macro F1-Score.

The Personalized Math Learning Engine was assessed through the accuracy improvement, mastery progression and error reduction measures, among others, in terms of learner performance and mastery progression.

The Adaptive Game-Based Learning Engine was assessed using measures of learner performance and engagement, such as accuracy, response time, progression, task completion and engagement.

The K-Means clustering and the Gaussian Mixture Model (GMM) were assessed for the Learning Insight and Early Warning system using the Silhouette Score and Davies–Bouldin Index. All the models were assessed by applying the following parameters: Accuracy, Precision, Recall, Macro F1-Score, Confusion Matrix and AUC-ROC on the Logistic Regression learning-risk prediction model, as well as on the benchmark models.

IV. Results And Discussion

The proposed dyscalculia support system provides a comprehensive, technology-driven approach to identifying and supporting students with mathematical learning difficulties. It assesses key cognitive skills, including memory, attention, number sense, and processing speed, while providing personalized learning and adaptive game-based activities. The system also monitors student performance and behavior to generate insights and identify students who may need early intervention.

The overall system integrates four key components: the Cognitive Skill Profiling System, Personalized Math Learning Engine, Adaptive Game-Based Learning Engine, and Learning Insight and Early Warning System. Together, these components support a continuous learning process, starting with cognitive assessment, followed by personalized and adaptive learning activities, while continuously monitoring student progress and potential learning difficulties. The overall evaluation demonstrates how these components work together to provide a cohesive learning support system for students with dyscalculia.

The proposed system consists of four main components, each using a suitable machine learning approach. The Cognitive Skill Profiling System selected Random Forest as its final model. The Personalized Math Learning Engine uses Bayesian Knowledge Tracing (BKT) to track learner progress, while the Adaptive Game-Based Learning Engine uses a rule-based model to adapt activities. Finally, the Learning Insight and Early Warning System uses K-Means and GMM for pattern analysis and Logistic Regression (Arm A) for learning-risk prediction.

To identify the most suitable approach for the Cognitive Skill Profiling System, several machine learning models were tested, including Decision Tree, Random Forest, Gradient Boosting, and Fine-Tuned Random Forest. The models were evaluated across four cognitive skills—Memory, Attention, Number Sense, and Processing Speed—using Macro F1-

score as the primary metric. Random Forest achieved the best overall performance with an Average Macro F1-score of 0.640 and was therefore selected as the final model. Although Fine-Tuned Random Forest performed better for Attention and Processing Speed individually, Random Forest provided the most balanced overall performance across all skills.

The Personalized Math Learning Engine continuously estimates learner knowledge and adapts mathematical activities based on individual progress. Bayesian Knowledge Tracing (BKT) estimates skill mastery from successive learner responses, providing advanced activities for higher mastery levels and additional practice for lower levels. The system also provides context-aware hints after incorrect responses. Its effectiveness is evaluated using accuracy improvement, skill mastery progression, repeated-error reduction, task completion, and learning time, with a target of at least 20% improvement in skill mastery.

To identify the most suitable approach for the Adaptive Game-Based Learning Engine, a Rule-Based Model and Reinforcement Learning were evaluated using Completion, Challenge-Fit, Engagement, and Safety & Stability. The Rule-Based Model showed better overall performance, while Reinforcement Learning performed slightly better in challenge-fit. Based on the results, the most suitable approach was selected to support adaptive game difficulty and learner engagement through points, rewards, badges, levels, and progression.

The Learning Insight and Early Warning System monitors learner performance and behavior across multiple learning sessions. It analyzes data such as performance, response time, attempts, errors, mastery, and engagement to identify learner patterns and students at risk of learning difficulties. Machine learning techniques are used to provide early intervention insights for teachers and parents, with performance evaluated using appropriate clustering and classification metrics.

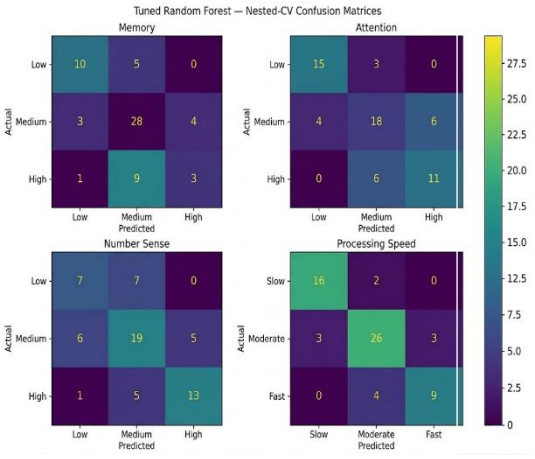


Figure 1 Confusion Matrix: Cognitive Skill Profiling System

The four components operate as a continuous learning cycle rather than independent applications. Student interactions, including answers, response time, attempts, and errors, are stored in the database. This data is used by the Personalized Math Learning Engine to update learner knowledge using BKT, while the Adaptive Game-Based Learning Engine adjusts gameplay based on performance and engagement. The accumulated data is then analyzed by the Learning Insight and Early Warning System to identify long-term learning patterns.

The integration of these components provides an advantage over conventional mathematics learning applications that mainly deliver exercises and record scores. In the proposed system, learner behavior continuously guides learning decisions. Repeated errors or slow responses trigger additional support, while improved mastery leads to more challenging activities. Long-term learning patterns are also analyzed to identify persistent difficulties and enable early intervention.

The overall system evaluation is expected to demonstrate whether the integration of cognitive profiling, personalized learning, adaptive gameplay, and learning analytics can improve the learning experience of children with dyscalculia.

For the overall evaluation, classification components are assessed using Accuracy, Precision, Recall, and F1-score, while learning-oriented components are evaluated using accuracy improvement, mastery progression, error reduction, response time, task completion, and engagement. Final results will be reported after evaluating the integrated system using collected learner data.

Module	Model	Memory Macro F1	Attention Macro F1	Number Sense Macro F1	Processing Speed Macro F1	Average Macro F1
Cognitive Skill Profiling System *(Random Forest Selected)*						
Cognitive Skill Profiling	Random Forest ✓	0.584	0.627	0.604	0.744	0.64
Cognitive Skill Profiling	Decision Tree Baseline	0.478	0.555	0.605	0.739	0.594
Cognitive Skill Profiling	Gradient Boosting	0.486	0.645	0.515	0.771	0.604
Cognitive Skill Profiling	Fine-Tuned Winner (RF)	0.525	0.658	0.567	0.793	0.636

Module	Model	Completion	Challenge-Fit	Engagement	Safety & Stability
Adaptive Game-Based Learning Engine *(Rule-Based Model selected)*					
Adaptive Game-Based Learning Engine	Rule-Based Model ✓	0.556	0.612	0.588	0.724
Adaptive Game-Based Learning Engine	Reinforcement Learning	0.473	0.632	0.503	0.635

Module	Model	Macro F1	Balanced Accuracy	F2 Score
Learning Insight & Early Warning System *(Logistic Regression (Arm A))*				
Learning Insight & Early Warning System	Logistic Regression (Arm A) ✓	0.424	0.477	0.689
Learning Insight & Early Warning System	Random Forest	0.426	0.479	0.729
Learning Insight & Early Warning System	Decision Tree	0.368	0.381	0.570
Learning Insight & Early Warning System	Logistic Regression (Arm B)	0.345	0.400	0.641
Learning Insight & Early Warning System	Time-bin Logistic Regression	0.323	0.423	0.662
Learning Insight & Early Warning System	Stratified Dummy Baseline	0.338	0.338	0.564
Learning Insight & Early Warning System	Majority-Class Baseline	0.230	0.333	0.877

Figure 2 Model Comparison and Benchmarking Across All Modules

To ensure consistent evaluation, each component was assessed using task-specific metrics. The Cognitive Skill Profiling System used Macro F1-score, while the Personalized Math Learning Engine measured mastery improvement, accuracy, error reduction, task completion, and learning time. The Adaptive Game-Based Learning Engine evaluated completion, challenge-fit, engagement, and stability. Finally, the Learning Insight and Early Warning System used Macro

F1, Balanced Accuracy, and F2-score for learning-risk prediction.

The benchmarking results are presented for each component, with the final model identified. Random Forest was selected for the Cognitive Skill Profiling System with an Average Macro F1-score of 0.640, while BKT was used for personalized knowledge estimation and learning adaptation, targeting at least 20% improvement in skill mastery. The Adaptive Game-Based Learning Engine compared Reinforcement Learning with a Rule-Based model, while Logistic Regression (Arm A) was selected for early learning-risk identification. Overall, the results show that selecting and optimizing different machine learning techniques for each component improves the effectiveness of the integrated dyscalculia learning support system.

Figure 3 Model Comparison and Benchmarking Across All Modules

V. CONCLUSION

This study introduces an Integrated Interactive Learning Tool for Children with Dyscalculia (IILT-DC) that combines cognitive skill profiling, personalized mathematics learning, adaptive game-based learning, and learning analytics in a single platform. Using learner performance, behavior, and engagement data, the system provides personalized learning activities, adaptive gameplay, and continuous progress tracking. It aims to create an engaging, learner-centered environment while supporting teachers and parents in understanding each learner's progress. Future work will involve testing the system with real learners, improving the machine learning models, and evaluating its effectiveness in real educational setting.

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